
Compositional Dynamics-Aligned Initial-Condition Codes for Physics-Informed Parametric Surrogates

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Abstract

Parameterized physics-informed neural networks amortize a PDE solve across equation coefficients, yet normally bake the initial condition into the loss. They therefore cannot be queried at a new initial function without retraining. We study whether a single physics-informed surrogate can generalize jointly over a coefficient and a function-valued initial condition on nonlinear reaction dynamics. Our method, compositional dynamics-aligned conditioning (CoDA), represents the initial condition by Fourier coefficients of its logit field and places independently encoded coefficient and function factors in anchored, disjoint latent subspaces. The construction reflects the exact identity $\text{logit } u(x, t) = \text{logit } u_0(x) + \rho t$ while retaining the coordinate encoder, decoder, physics loss, and training budget of P²INN. On a fully untouched three-seed confirmation, CoDA reaches relative L_2 errors of 0.1493, 0.1449, and 0.0137 on new Gaussian parameters, a new circular-logistic family, and withheld coefficient–initial-condition pairings. The matched raw-sensor encoder obtains 0.1977, 0.1917, and 0.1000, respectively. The frozen decision rule, conditioning interventions, and latent-geometry diagnostics separate representation effects from extra capacity. A sharpness sweep also shows where zero-shot transfer degrades. We deliberately make no claim about varying boundary conditions or PDE families: all experiments use one nonlinear reaction family with a fixed periodic boundary.

1 Introduction

Many-query PDE workflows vary more than material or transport coefficients. Sources, initial states, and boundary data are themselves functions. A surrogate conditioned only on coefficients is therefore not a surrogate over the full problem specification. This distinction is especially consequential for physics-informed neural networks (PINNs) [Raissi et al., 2019]: although initial and boundary conditions constrain training, they need not be inputs to the trained predictor.

P²INN [Cho et al., 2024] makes coefficient variation explicit. A coefficient encoder and a coordinate encoder produce separate 50-dimensional codes; a shared decoder maps their concatenation to the solution. In the reference implementation, however, the coefficient encoder receives only $\mu = (\beta, \nu, \rho)$. The initial condition is enforced at $t = 0$ and is fixed for the life of the model. Simply appending samples of u_0 to μ removes this interface limitation, but it does not say which function representation will extrapolate, nor whether a learned model actually uses that representation.

We isolate those questions on the periodic reaction family. It is small enough to support exhaustive full-grid evaluation against an analytic reference, but its map from u_0 to $u(t)$ is nonlinear; success cannot be explained by free linear superposition. We introduce CoDA, a surgical replacement for the P²INN parameter encoder. Its two inductive biases are (i) a periodic Fourier representation after the coordinate transform that linearizes the reaction flow, and (ii) anchored, disjoint subspaces for

coefficient and initial-condition codes. The latter makes unseen coefficient–condition composition an architectural property rather than a pattern a flat multilayer perceptron must infer.

Our evidence is organized around three questions.

- *Generalization*: can one final checkpoint handle unseen parameters of the training IC family, an entirely new IC family, and ICs paired with coefficients that were withheld during training?
- *Attribution*: does structured conditioning beat a parameter-matched encoder of raw function samples, without losing accuracy on the original fixed-IC task, and does replacing its code change outputs?
- *Mechanism and boundary*: does latent distance reflect a physical function-space distance, and does that distance predict error as IC shape sharpness moves out of distribution?

The experimental protocol was frozen before each comparison. Two completed development studies exposed non-seed-separated pairing results; we retained both, screened one architectural revision under a written advance rule, and only then froze new training and evaluation profiles. This separation matters: only the third, untouched split supports the headline conclusions.

2 Background and related work

Physics-informed parametric surrogates. A PINN represents a solution by a coordinate network and minimizes residual, initial, and boundary losses [Raissi et al., 2019, Karniadakis et al., 2021]. Training can be fragile for stiff or high-frequency regimes [Krishnapriyan et al., 2021]. Meta-initialization and hypernetworks amortize optimization across tasks [Liu et al., 2022, Cho et al., 2023], whereas P²INN explicitly embeds PDE coefficients and decodes many coefficient settings with one jointly trained model [Cho et al., 2024]. Our question is orthogonal: can that coefficient latent become a queryable coefficient–function latent while the rest of the P²INN is held fixed?

Operator learning. DeepONet [Lu et al., 2021], the Fourier neural operator [Li et al., 2021], and neural operators more generally [Kovachki et al., 2023] learn mappings between function spaces. Physics-informed DeepONets [Wang et al., 2021] and PINO [Li et al., 2022] incorporate governing equations; MIONet explicitly handles multiple input functions [Jin et al., 2022]. Most directly, PINTO uses cross-attention to condition a physics-only transformer operator on initial and boundary data and demonstrates transfer to unseen conditions [Boya and Subramani, 2025]. HyPINO spans operators, sources, geometries, and mixed boundary conditions by using a transformer hypernetwork with manufactured and physics-supervised samples [Bischof et al., 2025]. These works establish that function-conditioned, and even multi-physics, solution operators are viable. We do not claim to replace them. Instead, we conduct a controlled representation study inside the much smaller P²INN factorization: no solution trajectories are training labels, and the decoder, coordinate path, collocation, objective, and conditioned-model parameter count are identical across arms. Unlike PINTO’s learned integral kernel or HyPINO’s generated target networks, our central object is an explicit dynamics-aligned function coordinate and a separable coefficient–IC latent. This isolates representation from changing the surrogate class.

Conditional neural fields. Coordinate networks can represent individual signals continuously [Sitzmann et al., 2020], and Functa associates signals with latent modulations of a shared coordinate network [Dupont et al., 2022]. Such instance codes motivate treating an IC as a function rather than a list of task identifiers. Our code is produced directly from the queried function—no per-IC latent optimization—and its transform follows a known symmetry of the dynamics.

3 From coefficient codes to compositional IC codes

3.1 Held-fixed physics-informed surrogate

We use the reaction specialization of the supplied convection–diffusion– reaction benchmark on $x \in [0, 2\pi)$ and $t \in [0, 1]$:

$$u_t - \rho u(1 - u) = 0, \quad u(x, 0) = u_0(x), \quad u(0, t) = u(2\pi, t). \quad (1)$$

For a specification $s = (\boldsymbol{\mu}, u_0)$ with $\boldsymbol{\mu} = (0, 0, \rho)$, the reference P²INN computes

$$h_c = g_c(x, t), \quad h_s = g_p(\boldsymbol{\mu}), \quad \hat{u} = g_d([h_s; h_c], \boldsymbol{\mu}, x, t). \quad (2)$$

The last arguments denote the reference linear modulation path. We retain its three-layer coordinate encoder, 50+50 concatenation, six-layer residual decoder, modulation, activations, and automatic-differentiation residual exactly. We change only h_s .

All arms minimize the supplied equal-weight objective

$$\mathcal{L} = \mathcal{L}_f + \mathcal{L}_{f,0} + \mathcal{L}_{u_0} + \mathcal{L}_{\text{per}}, \quad (3)$$

where the first two terms enforce (1) at interior and IC points, the third matches $\hat{u}(x, 0)$ to the queried u_0 , and the last enforces periodic endpoint values. An IC appears both in the code and in its constraint; at test time only the code is needed.

3.2 Dynamics-aligned periodic function representation

For $0 < u < 1$, define $z = \text{logit}(u) = \log(u/(1-u))$. Equation (1) then gives

$$z_t = \frac{u_t}{u(1-u)} = \rho, \quad z(x, t) = z_0(x) + \rho t. \quad (4)$$

Thus coefficient and initial function act additively in an exact coordinate, even though $u_0 \mapsto u(t)$ is nonlinear. We sample u_0 at 64 uniform periodic sensors, clip only for numerical stability to $[10^{-4}, 1 - 10^{-4}]$, take z_0 , and pack the real Fourier coefficients through mode 16 as

$$\phi(u_0) = (a_0, a_1, b_1, \dots, a_{16}, b_{16}, 0, \dots, 0) \in \mathbb{R}^{64}. \quad (5)$$

The vector is divided by the largest retained absolute coefficient over training ICs only. Candidate cutoffs $\{8, 12, 16\}$ and a reconstruction criterion were fixed in the development plan. None met the stringent threshold on the clipped anchor, so the predeclared fallback $K = 16$ was used; its worst training-IC reconstruction RMS is 0.0094. The cutoff was then frozen for confirmation.

3.3 Anchored disjoint fusion at matched capacity

The CODA specification encoder is

$$\begin{aligned} h_\mu &= P_\mu \mu + 0.1 \tanh E_\mu(\mu) \in \mathbb{R}^9, \\ h_0 &= P_0 \phi(u_0) + 0.1 \tanh E_0(\phi(u_0)) \in \mathbb{R}^{41}, \\ h_s &= s \odot [h_\mu; h_0] + b. \end{aligned} \quad (6)$$

Here $P_\mu \mu$ places $\mu/5$ in the first three of nine coordinates and P_0 selects the first 41 IC-code coordinates. These fixed paths anchor the representation to interpretable inputs; bounded learned residuals retain flexibility. The tanh MLP widths are $E_\mu : 3 \rightarrow 138 \rightarrow 138 \rightarrow 138 \rightarrow 9$ and $E_0 : 64 \rightarrow 83 \rightarrow 83 \rightarrow 83 \rightarrow 41$; s and b are learned 50-vectors. The disjoint subspaces prevent an arbitrary learned rotation from entangling an IC with the coefficients it happened to accompany. This encoder has 63,050 parameters. Its raw-sensor comparator concatenates μ with $[2u_0(x_j) - 1]_{j=1}^{64}$ and uses the reference flat widths $67 \rightarrow 150 \rightarrow 150 \rightarrow 150 \rightarrow 50$, also exactly 63,050 parameters. Both complete models have 86,451 parameters. The coefficient-only model retains the original $3 \rightarrow 150 \rightarrow 150 \rightarrow 150 \rightarrow 50$ encoder and has 76,851 parameters.

Equation (6) obeys the rectangle identity

$$h_s(\mu_a, u_a) + h_s(\mu_b, u_b) - h_s(\mu_a, u_b) - h_s(\mu_b, u_a) = 0. \quad (7)$$

This is not an assertion that the PDE solution is additive. It is an inductive bias that the two specification factors remain independently addressable before their joint decoding, aligned with (4).

4 Experimental protocol

Training specifications. The final-confirmation training family contains the supplied Gaussian anchor and four wrapped Gaussians varying base, amplitude, width, and center. We train at $\rho \in \{1, 2, 3, 4, 5\}$. The anchor occurs at every coefficient; Gaussian f_i occurs iff $(i + \rho) \bmod 2 = 0$. This gives 15 unique specifications. A second anchor copy at each ρ balances the 20 training rows to four per coefficient. Figure 1 and Appendix A give all functions and pairs.

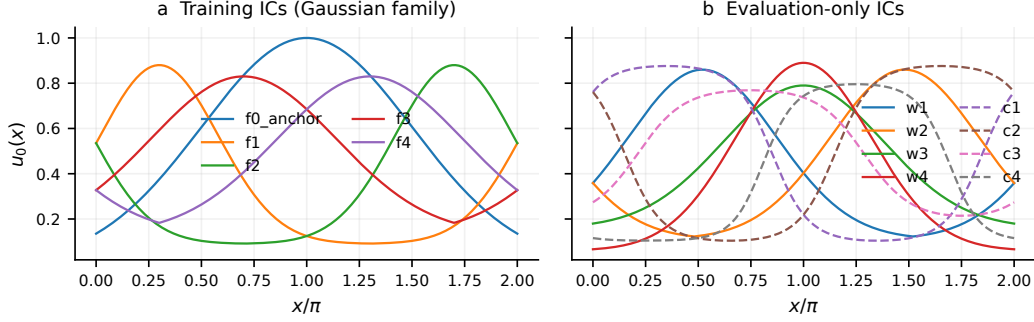


Figure 1: Every initial condition in the untouched confirmation. Solid evaluation curves are new parameters of the Gaussian family; dashed curves are a new circular-logistic family. None of the right-panel functions enters training or representation selection.

Evaluation-only specifications. Three core partitions test distinct extrapolations: (i) four new wrapped Gaussians at $\rho \in \{1, 3, 5\}$; (ii) four positive circular-logistic functions, a family absent from all development and final-confirmation training, at the same coefficients; and (iii) all ten Gaussian-coefficient pairings excluded by the checkerboard rule. Descriptive partitions contain the original anchor, training specifications, coefficients $\rho \in \{1.5, 2.5, 3.5, 4.5\}$ crossed with three ICs, and circular-logistic profiles with sharpness $\kappa = 1, \dots, 12$ at $\rho = 3$. No evaluation-only function or pairing is used to tune a representation or choose a checkpoint. The evaluated breadth is two IC families (Gaussian and circular-logistic), one boundary type, and one PDE family. Because the BC set is the singleton periodic condition, no BC descriptor is supplied and no BC-transfer result is implied.

Budget and metrics. Every row contributes 1,000 fixed random interior points, 256 IC points, and 100 boundary times, for 20,000, 5,120, and 2,000 points per epoch. All loss weights are one. We use full-batch Adam at 10^{-3} for 20,000 epochs and seeds 0, 1, and 2; reported predictions come from the final checkpoint. The two conditioned models’ seed runs use the same float32 data, optimizer, and single NVIDIA H200. We evaluate each specification on a 256×100 grid. We report relative L_2 , absolute RMS L_2 , and MAE, first averaged over the partition and then as mean $\pm$ sample SD over seeds.

Before running any comparison we set a practical margin $\delta = 0.010$ relative L_2 . A *clear* core win must beat a comparator by at least δ and by more than the pooled seed SD $\sqrt{(s_1^2 + s_2^2)}/2$. With only three deterministic seeds, we use this transparent effect-size rule rather than a fragile asymptotic p -value. Original-task non-regression is checked separately at every ρ : the allowed increase over the coefficient-only anchor error is $\max(0.1e_{\text{base}}, 0.005)$.

Reference and baseline parity. The exact solution is the supplied logistic construction generalized only by passing a new u_0 :

$$u(x, t) = \frac{u_0(x)e^{\rho t}}{1 - u_0(x) + u_0(x)e^{\rho t}}. \quad (8)$$

Our evaluator agrees with the supplied `reaction_solution` to 0.0e+00 maximum absolute error on the original anchor and has analytic ODE residual at most 6.66e-16. All new ICs are periodic. The coefficient-only checkpoints from development are reused in the final confirmation because that arm replaces every row’s IC with the same supplied anchor: coefficient multiset, coordinates, objective, architecture, epochs, and seeds are byte-for-byte equivalent inputs. This reuse was specified before confirmation, and checkpoint hashes are recorded.

5 Results

Table 1: Untouched confirmation on the nonlinear reaction family with a fixed periodic boundary. Each cell is relative L_2 (absolute RMS L_2 underneath), mean $\pm$ sample SD over three seeds; metrics are first averaged over every specification in the partition. Every arm has matched optimization, collocation, epoch, and seed budgets; conditioned models also have exactly matched size (86,451 parameters).

Method (parameters)	Seen	New Gaussian	New family	New pairing
Coefficient-only P ² INN	0.3016 $\pm$ 0.0004	0.3275 $\pm$ 0.0009	0.4142 $\pm$ 0.0005	0.4511 $\pm$ 0.0006
76,851	0.2014 $\pm$ 0.0003 abs.	0.2220 $\pm$ 0.0006 abs.	0.2946 $\pm$ 0.0004 abs.	0.3010 $\pm$ 0.0004 abs.
Raw-sensor encoder	0.0071 $\pm$ 0.0029	0.1977 $\pm$ 0.0288	0.1917 $\pm$ 0.0311	0.1000 $\pm$ 0.0258
86,451	0.0052 $\pm$ 0.0021 abs.	0.1327 $\pm$ 0.0175 abs.	0.1325 $\pm$ 0.0231 abs.	0.0667 $\pm$ 0.0183 abs.
CoDA (ours)	0.0047 $\pm$ 0.0017	0.1493 $\pm$ 0.0045	0.1449 $\pm$ 0.0029	0.0137 $\pm$ 0.0018
86,451	0.0034 $\pm$ 0.0013 abs.	0.1011 $\pm$ 0.0029 abs.	0.1018 $\pm$ 0.0020 abs.	0.0090 $\pm$ 0.0011 abs.

Table 2: Predeclared confirmation decisions against both comparators. A clear win requires a relative- L_2 margin ≥ 0.010 and larger than the pooled seed SD.

Axis	Comparator	Margin	Pooled SD	Paired SD	Clear
New Gaussian	Raw sensors	0.0484	0.0206	0.0302	yes
New Gaussian	Coefficient-only	0.1782	0.0033	0.0037	yes
New family	Raw sensors	0.0468	0.0221	0.0292	yes
New family	Coefficient-only	0.2693	0.0021	0.0023	yes
New pairing	Raw sensors	0.0864	0.0183	0.0265	yes
New pairing	Coefficient-only	0.4374	0.0013	0.0020	yes

5.1 Joint coefficient-IC generalization

Table 1 reports the untouched confirmation. On new Gaussian parameters, CoDA reduces relative error by 24.5% versus raw sensors, from 0.1977 ± 0.0288 to 0.1493 ± 0.0045 . On the wholly new circular-logistic family, the reduction is 24.4%, and on withheld coefficient-IC pairings it is 86.3%. The corresponding raw-control margins are 0.0484, 0.0468, and 0.0864. Table 2 compares each with the predeclared tolerance and seed variation; the coefficient-only model is also worse on every variable-IC partition. Complete absolute errors, per-seed values, and per-specification results appear in Appendix B. Figure 6 exposes every constituent comparison rather than relying only on partition averages. The improvement is heterogeneous rather than universal: CoDA has lower three-seed mean error than raw sensors on 10/ 12 new-Gaussian cases and 8/12 new-family cases, while doing so on 10/10 withheld pairings.

The seen result is not purchased by underfitting: CoDA obtains 0.0047 ± 0.0017 on the 15 training specifications, versus 0.0071 ± 0.0029 for raw sensors. More importantly, the original fixed-IC task remains intact. Anchor error is 0.0037 ± 0.0016 compared with 0.0032 ± 0.0015 for the reproduced coefficient-only model, and every coefficient passes the frozen non-regression tolerance (Appendix C).

5.2 OOD behavior and failure analysis

Coefficient interpolation was not used for the decision. CoDA reaches 0.0776 ± 0.0089 , compared with 0.1286 ± 0.0400 for raw sensors and 0.3209 ± 0.0012 for the coefficient-only model. Figure 2b pushes a new circular-logistic IC from sharpness 1 through 12. At $\kappa = 2$, CoDA has relative error 0.0607 ± 0.0168 ; at $\kappa = 12$ it is 0.2245 ± 0.0132 . In this far tail the raw encoder reaches 0.1788 ± 0.0128 , so the structured advantage has reversed: the dynamics-aligned coordinate alone does not guarantee decoder extrapolation to increasingly sharp shapes. This is a useful empirical failure boundary, not evidence of a fundamental impossibility: we did not run the required capacity and collocation scaling study. Interestingly, input reconstruction at $\kappa = 12$ remains 0.0003 RMS, so output error cannot be attributed solely to loss of that IC in the Fourier code; decoder extrapolation and training support also matter.

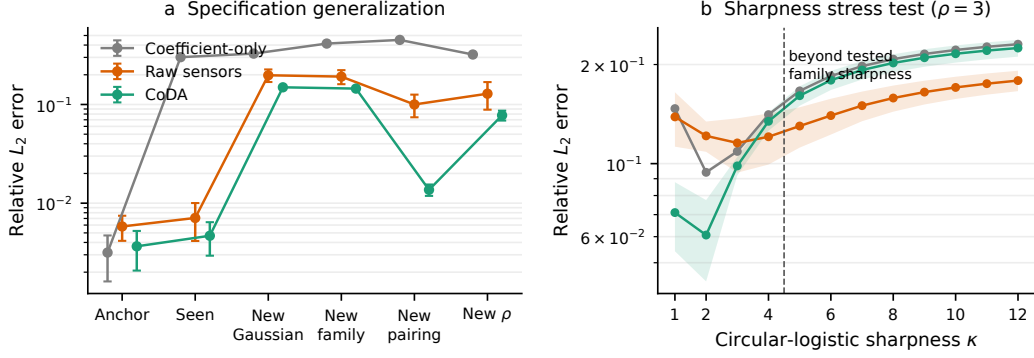


Figure 2: Mean relative L_2 with one seed SD. (a) All principal confirmation partitions. (b) A declared shape-sharpness stress test on a circular-logistic IC; the dashed line separates sharpness represented in the tested new-family profiles ($\kappa \leq 4.2$) from further extrapolation. Every point aggregates all three final checkpoints.

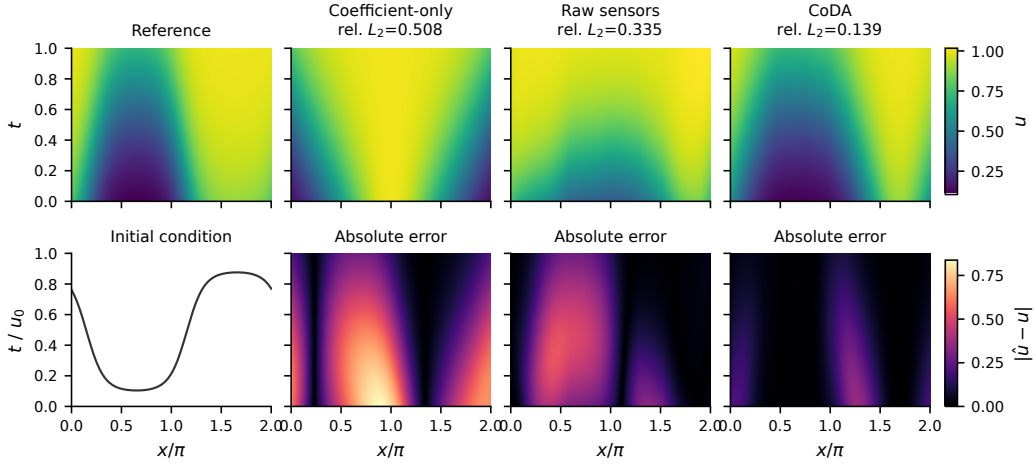


Figure 3: Seed-0 solution fields for the fixed illustrative new-family case c_2 at $\rho = 3$ (selected by index, not by error), with absolute-error maps below. The first lower panel shows its initial condition.

5.3 Does the network use the IC code?

At inference, we intervene on CODA without retraining. The *mean-code* intervention replaces each IC representation by the mean over the five training ICs. The *deranged-code* intervention cyclically permutes target IC codes within each ρ , so the coefficient and all network weights remain correct. Across the three core held-out partitions together, these interventions multiply error by 3.1 and 4.1, respectively. On new Gaussians the factors are 1.8 and 2.6; on withheld pairings they are 27.7 and 29.6. The full partition breakdown in Table 3 prevents a favorable aggregate from hiding a weak axis. All six held-out axis/intervention degradations exceed their pooled seed SD, with the smallest degradation-to-noise ratio $19.5\times$. The large seen and pairing changes rule out an ignored or constant encoding.

6 What is represented?

We examine all five training ICs, four new Gaussians, and four new-family ICs at fixed $\rho = 3$. For each seed, we compare all pairwise Euclidean distances between 50-dimensional specification codes with RMS distances between the corresponding logit fields on 256 points. Spearman correlation is 0.933 ± 0.008 across seeds. The minimum pairwise latent distance over all seeds is 0.141, so distinct

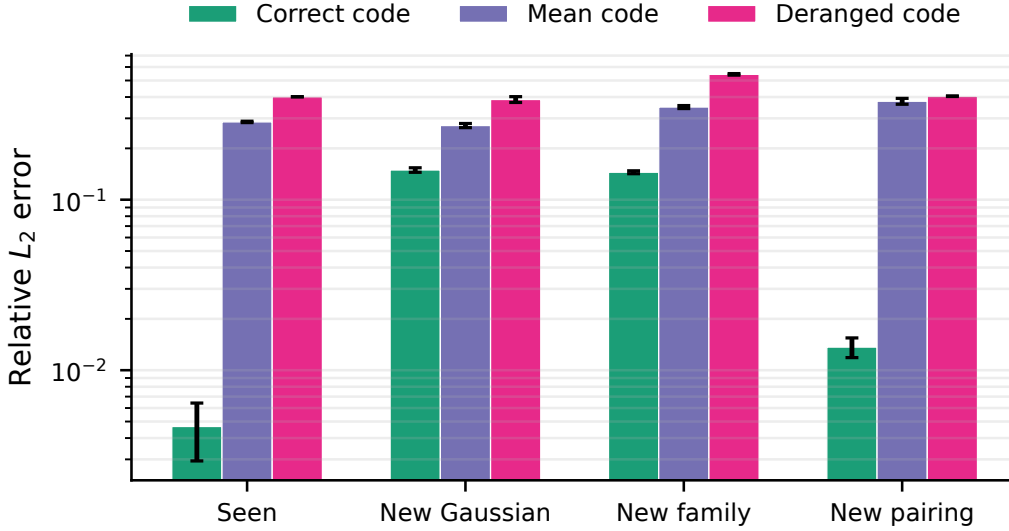


Figure 4: IC-code interventions. Bars are mean relative L_2 and one seed SD; the vertical axis is logarithmic.

Table 3: Conditioning interventions (relative L_2 , mean $\pm$ seed SD). Predictions and targets are unchanged; only the IC code is replaced.

Code	Seen	New Gauss.	New family	New pairing
Correct IC code	0.0047 $\pm$ 0.0017	0.1493 $\pm$ 0.0045	0.1449 $\pm$ 0.0029	0.0137 $\pm$ 0.0018
Mean training code	0.2859 $\pm$ 0.0027	0.2722 $\pm$ 0.0076	0.3494 $\pm$ 0.0070	0.3778 $\pm$ 0.0150
Within- ρ derangement	0.4014 $\pm$ 0.0004	0.3869 $\pm$ 0.0151	0.5432 $\pm$ 0.0056	0.4044 $\pm$ 0.0016

ICs have not collapsed. The seed-0 PCA in Figure 5 shows new Gaussians extending the training geometry while the new functional family occupies a distinct region.

For a diagnostic not used in selection, define IC novelty as distance to the nearest training logit field. Across every new-Gaussian, new-family, and sharpness-sweep specification, novelty and mean CODA error have Spearman $r_s = 0.540$ ($p = 6.69e - 04$). This connects the code’s physical geometry to its observed transfer boundary rather than merely visualizing clusters. Finally, the learned encoder satisfies the architectural rectangle law (7) to a measured maximum relative residual of $4.31e-08$, while the analytic logit-flow identity has maximum RMS residual $1.23e-11$. Geometry, intervention, and generalization thus support complementary claims: the code varies with the IC, carries a dynamics-relevant metric, and remains compositionally separable from ρ .

7 Limitations and conclusion

The study answers a narrow question cleanly. It covers one nonlinear reaction PDE, one periodic boundary type, five training ICs from one family, and three random seeds. It does not establish boundary-condition transfer, joint multi-PDE training, high-dimensional scalability, or superiority to modern neural operators. The logit transform uses known reaction structure and requires bounded ICs; a different dynamics family may need a different coordinate or a learned invertible representation. Fourier truncation is natural on a periodic domain but not on an irregular geometry. Our effect-size rule exposes seed noise but three seeds do not support precise tail-probability claims. The sharpness sweep maps degradation only at the fixed budget, so we do not call it fundamental.

Within that scope, the result changes what the compact P²INN surrogate can be queried on. An initial condition that previously existed only in the loss becomes an inference-time function input. Dynamics-aligned representation and anchored, disjoint coefficient–IC fusion together outperform

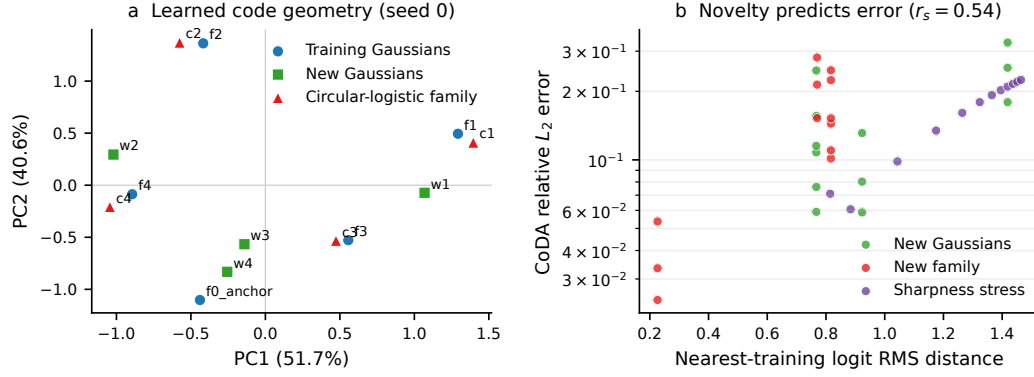


Figure 5: Representation diagnostics. (a) PCA of seed-0 codes at fixed $\rho = 3$. (b) Error versus nearest-training distance in the physical logit metric; repeated profile points reflect different coefficients. The correlation is descriptive and is computed over all plotted specifications.

a matched raw-input path across three fresh generalization axes while retaining the original anchor accuracy. The broader lesson is not that one logit-Fourier code is universal, but that symmetry and factorization in specification space can be as important as the coordinate architecture used to decode a solution.

Broader impact. Amortizing multiple solves can reduce repeated training cost in simulation workflows. Conversely, a zero-shot surrogate may be trusted outside its validated specification region; Figure 2 shows why distance-aware uncertainty or fallback solvers remain necessary in safety-critical use.

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A Exact profiles, splits, and budgets

Frozen artifacts. The development plan was frozen at 2026-09-05T09:59:23+09:00 (SHA-256 prefix 7436a876309a). After that study, the additive architecture and an entirely fresh second split were frozen at 2026-09-05T11:05:37+09:00 (prefix 1d0b30302d64). Its pairing result was again inconclusive. The anchored candidate screen was frozen at 2026-09-05T11:51:44+09:00 (prefix d37510d17329). After it passed its written one-seed rule, the unchanged method and third split used for every headline claim were frozen at 2026-09-05T12:00:44+09:00 (prefix 6321be056046). The machine-readable audit verifies that each timestamp precedes its governed runs and records every complete SHA-256; the prefixes here distinguish the four preregistrations without disrupting typesetting.

Let $d(x, c) = \min(|x - c|, 2\pi - |x - c|)$ for the wrapped profiles. Final confirmation training uses the supplied $f_0(x) = \exp[-\frac{1}{2}((x - \pi)/(\pi/2))^2]$ and

$$\begin{aligned} f_1 &= .09 + .79e^{-\frac{1}{2}(d(x, .30\pi)/(.28\pi))^2}, & f_2 &= .09 + .79e^{-\frac{1}{2}(d(x, 1.70\pi)/(.28\pi))^2}, \\ f_3 &= .13 + .70e^{-\frac{1}{2}(d(x, .70\pi)/(.44\pi))^2}, & f_4 &= .13 + .70e^{-\frac{1}{2}(d(x, 1.30\pi)/(.44\pi))^2}. \end{aligned}$$

New-parameter ICs are

$$\begin{aligned} w_1 &= (.11, .75, .35\pi, .52\pi), & w_2 &= (.11, .75, .35\pi, 1.48\pi), \\ w_3 &= (.16, .63, .38\pi, 1.00\pi), & w_4 &= (.06, .83, .32\pi, 1.00\pi), \end{aligned}$$

where tuples denote (base, amplitude, width, center). New-family ICs have $c(x) = b + a\sigma(\kappa(\cos(x - m) - \tau))$. Their $(b, a, \kappa, \tau, m/\pi)$ tuples are $c_1 = (.08, .82, 3.5, 0, .35)$, $c_2 = (.08, .82, 3.5, 0, 1.65)$, $c_3 = (.15, .65, 2.6, -.15, .75)$, and $c_4 = (.10, .72, 4.2, .2, 1.25)$. The machine-readable plan enumerates every training and test pair. Boundary and PDE-family breadth are both exactly one; neither excluded stretch experiment is implied by the phrase “specification encoder.”

Table 4: Architectural and optimization parity.

	Coefficient-only	Raw sensors	CODA
Specification encoder parameters	53,450	63,050	63,050
Coordinate encoder parameters	5,250	5,250	5,250
Decoder and modulation parameters	18,151	18,151	18,151
Total parameters	76,851	86,451	86,451
Epochs	20,000	20,000	20,000
Interior / IC / boundary points	20,000 / 5,120 / 2,000	same	same
Optimizer / learning rate	Adam / 10^{-3}	same	same
Seeds	0, 1, 2	0, 1, 2	0, 1, 2

B Complete confirmation results

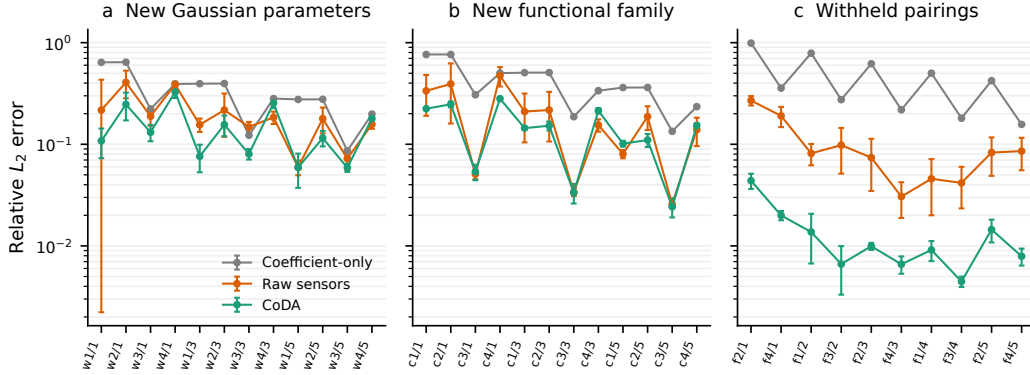


Figure 6: Every core confirmation specification, with mean relative L_2 and one seed SD. Labels are profile/coefficient; connecting lines aid tracing and do not imply continuity between profiles.

Table 5: Complete confirmation aggregates. Values are mean $\pm$ seed SD after averaging across all specifications in a partition.

Method	Partition	Relative L_2	Absolute RMS L_2	MAE
Coefficient-only P^2 INN	Original anchor	0.0032 $\pm$ 0.0015	0.0025 $\pm$ 0.0013	0.0019 $\pm$ 0.0009
Coefficient-only P^2 INN	Seen specifications	0.3016 $\pm$ 0.0004	0.2014 $\pm$ 0.0003	0.1665 $\pm$ 0.0003
Coefficient-only P^2 INN	New Gaussian parameters	0.3275 $\pm$ 0.0009	0.2220 $\pm$ 0.0006	0.1818 $\pm$ 0.0005
Coefficient-only P^2 INN	New circular-logistic family	0.4142 $\pm$ 0.0005	0.2946 $\pm$ 0.0004	0.2343 $\pm$ 0.0003
Coefficient-only P^2 INN	Withheld coefficient–IC pairs	0.4511 $\pm$ 0.0006	0.3010 $\pm$ 0.0004	0.2490 $\pm$ 0.0004
Coefficient-only P^2 INN	Interpolated coefficients	0.3209 $\pm$ 0.0012	0.2333 $\pm$ 0.0009	0.1900 $\pm$ 0.0006
Coefficient-only P^2 INN	Sharpness stress test	0.1784 $\pm$ 0.0009	0.1349 $\pm$ 0.0007	0.0989 $\pm$ 0.0005
Raw-sensor encoder	Original anchor	0.0058 $\pm$ 0.0017	0.0047 $\pm$ 0.0013	0.0042 $\pm$ 0.0016
Raw-sensor encoder	Seen specifications	0.0071 $\pm$ 0.0029	0.0052 $\pm$ 0.0021	0.0046 $\pm$ 0.0022
Raw-sensor encoder	New Gaussian parameters	0.1977 $\pm$ 0.0288	0.1327 $\pm$ 0.0175	0.1059 $\pm$ 0.0154
Raw-sensor encoder	New circular-logistic family	0.1917 $\pm$ 0.0311	0.1325 $\pm$ 0.0231	0.1007 $\pm$ 0.0175
Raw-sensor encoder	Withheld coefficient–IC pairs	0.1000 $\pm$ 0.0258	0.0667 $\pm$ 0.0183	0.0537 $\pm$ 0.0140
Raw-sensor encoder	Interpolated coefficients	0.1286 $\pm$ 0.0400	0.0926 $\pm$ 0.0281	0.0712 $\pm$ 0.0209
Raw-sensor encoder	Sharpness stress test	0.1470 $\pm$ 0.0140	0.1113 $\pm$ 0.0105	0.0839 $\pm$ 0.0069
CoDA (ours)	Original anchor	0.0037 $\pm$ 0.0016	0.0029 $\pm$ 0.0013	0.0021 $\pm$ 0.0009
CoDA (ours)	Seen specifications	0.0047 $\pm$ 0.0017	0.0034 $\pm$ 0.0013	0.0025 $\pm$ 0.0010
CoDA (ours)	New Gaussian parameters	0.1493 $\pm$ 0.0045	0.1011 $\pm$ 0.0029	0.0801 $\pm$ 0.0046
CoDA (ours)	New circular-logistic family	0.1449 $\pm$ 0.0029	0.1018 $\pm$ 0.0020	0.0722 $\pm$ 0.0012
CoDA (ours)	Withheld coefficient–IC pairs	0.0137 $\pm$ 0.0018	0.0090 $\pm$ 0.0011	0.0070 $\pm$ 0.0010
CoDA (ours)	Interpolated coefficients	0.0776 $\pm$ 0.0089	0.0568 $\pm$ 0.0065	0.0410 $\pm$ 0.0051
CoDA (ours)	Sharpness stress test	0.1641 $\pm$ 0.0079	0.1238 $\pm$ 0.0059	0.0933 $\pm$ 0.0030

Table 6: Seen coefficient–IC specifications: per-specification relative L_2 , mean $\pm$ seed SD.

IC	ρ	Coefficient-only	Raw sensors	CoDA
f0_anchor	1	0.0038 $\pm$ 0.0013	0.0072 $\pm$ 0.0021	0.0046 $\pm$ 0.0017
f1	1	0.9895 $\pm$ 0.0014	0.0103 $\pm$ 0.0048	0.0082 $\pm$ 0.0030
f3	1	0.3557 $\pm$ 0.0015	0.0095 $\pm$ 0.0040	0.0057 $\pm$ 0.0021
f0_anchor	2	0.0037 $\pm$ 0.0022	0.0066 $\pm$ 0.0017	0.0049 $\pm$ 0.0029
f2	2	0.7894 $\pm$ 0.0013	0.0077 $\pm$ 0.0036	0.0075 $\pm$ 0.0013
f4	2	0.2764 $\pm$ 0.0014	0.0069 $\pm$ 0.0022	0.0045 $\pm$ 0.0019
f0_anchor	3	0.0032 $\pm$ 0.0019	0.0056 $\pm$ 0.0019	0.0034 $\pm$ 0.0016
f1	3	0.6194 $\pm$ 0.0015	0.0094 $\pm$ 0.0065	0.0069 $\pm$ 0.0023
f3	3	0.2174 $\pm$ 0.0016	0.0070 $\pm$ 0.0031	0.0040 $\pm$ 0.0014
f0_anchor	4	0.0028 $\pm$ 0.0011	0.0050 $\pm$ 0.0014	0.0027 $\pm$ 0.0009
f2	4	0.5012 $\pm$ 0.0011	0.0074 $\pm$ 0.0044	0.0041 $\pm$ 0.0015
f4	4	0.1815 $\pm$ 0.0015	0.0056 $\pm$ 0.0022	0.0028 $\pm$ 0.0011
f0_anchor	5	0.0023 $\pm$ 0.0013	0.0047 $\pm$ 0.0013	0.0027 $\pm$ 0.0010
f1	5	0.4218 $\pm$ 0.0009	0.0081 $\pm$ 0.0052	0.0050 $\pm$ 0.0025
f3	5	0.1560 $\pm$ 0.0010	0.0053 $\pm$ 0.0021	0.0032 $\pm$ 0.0018

Table 7: Unseen Gaussian parameters: per-specification relative L_2 , mean $\pm$ seed SD.

IC	ρ	Coefficient-only	Raw sensors	CoDA
w1	1	0.6407 $\pm$ 0.0016	0.2170 $\pm$ 0.2148	0.1079 $\pm$ 0.0350
w2	1	0.6422 $\pm$ 0.0018	0.4068 $\pm$ 0.1235	0.2468 $\pm$ 0.0749
w3	1	0.2227 $\pm$ 0.0014	0.1887 $\pm$ 0.0364	0.1312 $\pm$ 0.0241
w4	1	0.3920 $\pm$ 0.0018	0.3870 $\pm$ 0.0253	0.3273 $\pm$ 0.0419
w1	3	0.3948 $\pm$ 0.0016	0.1557 $\pm$ 0.0233	0.0760 $\pm$ 0.0230
w2	3	0.3960 $\pm$ 0.0012	0.2173 $\pm$ 0.0989	0.1560 $\pm$ 0.0358
w3	3	0.1226 $\pm$ 0.0013	0.1484 $\pm$ 0.0177	0.0803 $\pm$ 0.0095
w4	3	0.2818 $\pm$ 0.0010	0.1840 $\pm$ 0.0255	0.2538 $\pm$ 0.0268
w1	5	0.2762 $\pm$ 0.0011	0.0593 $\pm$ 0.0097	0.0591 $\pm$ 0.0219
w2	5	0.2771 $\pm$ 0.0007	0.1787 $\pm$ 0.0509	0.1152 $\pm$ 0.0199
w3	5	0.0862 $\pm$ 0.0009	0.0723 $\pm$ 0.0165	0.0588 $\pm$ 0.0053
w4	5	0.1982 $\pm$ 0.0014	0.1572 $\pm$ 0.0152	0.1792 $\pm$ 0.0244

Table 8: Unseen circular-logistic family: per-specification relative L_2 , mean $\pm$ seed SD.

IC	ρ	Coefficient-only	Raw sensors	CoDA
c1	1	0.7650 $\pm$ 0.0009	0.3358 $\pm$ 0.1453	0.2239 $\pm$ 0.0049
c2	1	0.7661 $\pm$ 0.0016	0.3923 $\pm$ 0.2322	0.2473 $\pm$ 0.0165
c3	1	0.3071 $\pm$ 0.0016	0.0509 $\pm$ 0.0056	0.0536 $\pm$ 0.0094
c4	1	0.5012 $\pm$ 0.0013	0.4725 $\pm$ 0.1022	0.2813 $\pm$ 0.0111
c1	3	0.5069 $\pm$ 0.0012	0.2099 $\pm$ 0.1055	0.1442 $\pm$ 0.0041
c2	3	0.5077 $\pm$ 0.0011	0.2173 $\pm$ 0.1109	0.1522 $\pm$ 0.0151
c3	3	0.1868 $\pm$ 0.0016	0.0338 $\pm$ 0.0028	0.0334 $\pm$ 0.0073
c4	3	0.3368 $\pm$ 0.0009	0.1546 $\pm$ 0.0215	0.2140 $\pm$ 0.0130
c1	5	0.3615 $\pm$ 0.0006	0.0803 $\pm$ 0.0074	0.1014 $\pm$ 0.0066
c2	5	0.3621 $\pm$ 0.0007	0.1875 $\pm$ 0.0493	0.1101 $\pm$ 0.0160
c3	5	0.1341 $\pm$ 0.0011	0.0257 $\pm$ 0.0021	0.0243 $\pm$ 0.0052
c4	5	0.2347 $\pm$ 0.0005	0.1393 $\pm$ 0.0433	0.1525 $\pm$ 0.0092

Table 9: Withheld coefficient–IC pairings: per-specification relative L_2 , mean $\pm$ seed SD.

IC	ρ	Coefficient-only	Raw sensors	CoDA
f2	1	0.9908 $\pm$ 0.0016	0.2696 $\pm$ 0.0286	0.0438 $\pm$ 0.0075
f4	1	0.3573 $\pm$ 0.0018	0.1904 $\pm$ 0.0429	0.0200 $\pm$ 0.0021
f1	2	0.7882 $\pm$ 0.0014	0.0814 $\pm$ 0.0193	0.0137 $\pm$ 0.0070
f3	2	0.2749 $\pm$ 0.0015	0.0981 $\pm$ 0.0467	0.0066 $\pm$ 0.0033
f2	3	0.6203 $\pm$ 0.0012	0.0739 $\pm$ 0.0393	0.0099 $\pm$ 0.0008
f4	3	0.2187 $\pm$ 0.0012	0.0306 $\pm$ 0.0118	0.0066 $\pm$ 0.0013
f1	4	0.5006 $\pm$ 0.0007	0.0458 $\pm$ 0.0258	0.0091 $\pm$ 0.0020
f3	4	0.1806 $\pm$ 0.0006	0.0417 $\pm$ 0.0183	0.0045 $\pm$ 0.0005
f2	5	0.4225 $\pm$ 0.0005	0.0829 $\pm$ 0.0340	0.0145 $\pm$ 0.0037
f4	5	0.1570 $\pm$ 0.0009	0.0857 $\pm$ 0.0302	0.0079 $\pm$ 0.0015

Table 10: Coefficient interpolation: per-specification relative L_2 , mean $\pm$ seed SD.

IC	ρ	Coefficient-only	Raw sensors	CoDA
f0_anchor	1.5	0.0313 $\pm$ 0.0119	0.0359 $\pm$ 0.0139	0.0054 $\pm$ 0.0023
w1	1.5	0.5756 $\pm$ 0.0055	0.2211 $\pm$ 0.0894	0.1007 $\pm$ 0.0291
c1	1.5	0.7063 $\pm$ 0.0024	0.3213 $\pm$ 0.1716	0.2000 $\pm$ 0.0039
f0_anchor	2.5	0.0101 $\pm$ 0.0021	0.0146 $\pm$ 0.0053	0.0043 $\pm$ 0.0026
w1	2.5	0.4409 $\pm$ 0.0032	0.2347 $\pm$ 0.0428	0.0841 $\pm$ 0.0242
c1	2.5	0.5609 $\pm$ 0.0034	0.2460 $\pm$ 0.1492	0.1603 $\pm$ 0.0024
f0_anchor	3.5	0.0115 $\pm$ 0.0094	0.0076 $\pm$ 0.0009	0.0029 $\pm$ 0.0012
w1	3.5	0.3578 $\pm$ 0.0023	0.1053 $\pm$ 0.0320	0.0701 $\pm$ 0.0217
c1	3.5	0.4633 $\pm$ 0.0029	0.1720 $\pm$ 0.0650	0.1302 $\pm$ 0.0063
f0_anchor	4.5	0.0131 $\pm$ 0.0039	0.0098 $\pm$ 0.0013	0.0024 $\pm$ 0.0006
w1	4.5	0.2938 $\pm$ 0.0013	0.0796 $\pm$ 0.0154	0.0614 $\pm$ 0.0214
c1	4.5	0.3855 $\pm$ 0.0028	0.0949 $\pm$ 0.0066	0.1100 $\pm$ 0.0085

Table 11: Circular-logistic sharpness stress test: per-specification relative L_2 , mean $\pm$ seed SD.

IC	ρ	Coefficient-only	Raw sensors	CoDA
sharp_1	3	0.1470 $\pm$ 0.0011	0.1387 $\pm$ 0.0261	0.0710 $\pm$ 0.0168
sharp_2	3	0.0941 $\pm$ 0.0013	0.1214 $\pm$ 0.0128	0.0607 $\pm$ 0.0168
sharp_3	3	0.1089 $\pm$ 0.0011	0.1156 $\pm$ 0.0217	0.0984 $\pm$ 0.0064
sharp_4	3	0.1409 $\pm$ 0.0010	0.1207 $\pm$ 0.0213	0.1345 $\pm$ 0.0086
sharp_5	3	0.1663 $\pm$ 0.0009	0.1301 $\pm$ 0.0182	0.1608 $\pm$ 0.0109
sharp_6	3	0.1845 $\pm$ 0.0009	0.1400 $\pm$ 0.0175	0.1793 $\pm$ 0.0119
sharp_7	3	0.1978 $\pm$ 0.0008	0.1500 $\pm$ 0.0155	0.1925 $\pm$ 0.0124
sharp_8	3	0.2077 $\pm$ 0.0008	0.1582 $\pm$ 0.0144	0.2022 $\pm$ 0.0127
sharp_9	3	0.2153 $\pm$ 0.0008	0.1649 $\pm$ 0.0137	0.2097 $\pm$ 0.0128
sharp_10	3	0.2214 $\pm$ 0.0008	0.1704 $\pm$ 0.0132	0.2157 $\pm$ 0.0130
sharp_11	3	0.2264 $\pm$ 0.0008	0.1749 $\pm$ 0.0130	0.2205 $\pm$ 0.0131
sharp_12	3	0.2305 $\pm$ 0.0008	0.1788 $\pm$ 0.0128	0.2245 $\pm$ 0.0132

C Original-task non-regression

Table 12: Original-anchor non-regression by trained coefficient (relative L_2). The allowed increase is $\max(0.1e_{\text{base}}, 0.005)$.

ρ	Coefficient-only	CoDA	Increase	Pass
1	0.0038 $\pm$ 0.0013	0.0046 $\pm$ 0.0017	0.0007	yes
2	0.0037 $\pm$ 0.0022	0.0049 $\pm$ 0.0029	0.0012	yes
3	0.0032 $\pm$ 0.0019	0.0034 $\pm$ 0.0016	0.0002	yes
4	0.0028 $\pm$ 0.0011	0.0027 $\pm$ 0.0009	-7.14e-05	yes
5	0.0023 $\pm$ 0.0013	0.0027 $\pm$ 0.0010	0.0004	yes

The test compares the new model with independently trained coefficient-only checkpoints at each of the five original coefficients. It is stricter than reporting only a pooled anchor average: one coefficient cannot compensate for regression at another.

D Completed development studies

The first frozen method used the same logit–Fourier vector but concatenated it with μ before the flat $67 \rightarrow 150 \rightarrow 150 \rightarrow 150 \rightarrow 50$ encoder. It completed all nine runs. As Table 13 shows, it improved on raw sensors for new Gaussian parameters and a new sine family. Its new-pairing margin was 0.0118, above δ , but below the pooled seed SD 0.0362. We therefore did not declare success. Additive fusion was motivated by this failure and evaluated on a new three-seed split. It passed the new-parameter and new-family rules but again missed the pairing rule: its raw-control margin was 0.0030 versus pooled SD 0.0202. We next froze the anchored disjoint encoder and a one-seed advance rule. It obtained 0.0114 on the development pairing partition and 0.0052 on seen specifications, passing the respective < 0.060 and < 0.010 thresholds. Only then was the method held fixed and the final profile split created. Except for the mandatory supplied anchor, no development profile appears in final confirmation. All plans, checkpoints, and evaluations are retained.

Table 13: Completed development study (relative L_2 , mean $\pm$ seed SD). The flat encoder’s pairing margin was not seed-separated, motivating compositional fusion and an entirely fresh confirmation split. These data were not used as confirmatory evidence.

Method	Seen	New Gaussian	New sine family	New pairing
Coefficient-only P ² INN	0.2363 $\pm$ 0.0004	0.1827 $\pm$ 0.0008	0.3382 $\pm$ 0.0006	0.3681 $\pm$ 0.0008
Raw sensors	0.0037 $\pm$ 0.0004	0.1231 $\pm$ 0.0329	0.2412 $\pm$ 0.0168	0.1008 $\pm$ 0.0364
Flat dynamics-aligned	0.0067 $\pm$ 0.0010	0.0792 $\pm$ 0.0037	0.2009 $\pm$ 0.0242	0.0890 $\pm$ 0.0359

Table 14: Second development study and anchored screen (relative L_2). The first three rows are mean $\pm$ SD over three seeds. The final row is the predeclared one-seed architecture screen and is not inferential evidence.

Method	Seen	New Gaussian	New squared-sine	New pairing
Coefficient-only P ² INN	0.2642 $\pm$ 0.0003	0.2938 $\pm$ 0.0009	0.3541 $\pm$ 0.0004	0.3949 $\pm$ 0.0006
Raw sensors	0.0039 $\pm$ 0.0011	0.1997 $\pm$ 0.0261	0.3559 $\pm$ 0.0160	0.0818 $\pm$ 0.0138
Additive fusion	0.0036 $\pm$ 0.0003	0.1392 $\pm$ 0.0248	0.2723 $\pm$ 0.0038	0.0788 $\pm$ 0.0250
Anchored disjoint screen (seed 0)	0.0052	0.1069	0.2655	0.0114

E Reproducibility and artifact map

All code authored for this work is in `proposal/code/`; the supplied `code/P2INN/` tree is read-only. Each training run writes a JSON summary, a complete 500-epoch trace, and a final checkpoint. Each evaluation JSON contains every specification and all five raw metrics before aggregation. `final_aggregate_metrics.json` is the sole source for headline tables; `final_latent_diagnostics.json` and `reference_validation.json` source the mechanism and reference claims. `paper_numbers.tex` and every included table are generated from those JSON files by `make_paper_assets.py`. Figure values are additionally serialized in `figure_data.json`. The audit records all checkpoint SHA-256 hashes, exact split membership, plan timing, grid sizes, budgets, architecture parity, and final epochs.

The top-level reproduction script reruns selection, every study, aggregation, diagnostics, figures, integrity checks, and LaTeX compilation. Full training is deterministic conditional on the recorded PyTorch/CUDA stack and seeds; small GPU-kernel differences may remain across hardware. The submitted logs record PyTorch 2.13.0+cu126, float32 training, an NVIDIA H200, and analytic float64 references.

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1. **Claims.** Do the main claims accurately reflect the paper’s contributions and scope? [\[Yes\]](#) The abstract, results, and conclusion state only the axes tested on the untouched confirmation, and explicitly exclude BC and PDE-family transfer.
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4. **Experimental result reproducibility.** Is the information needed to reproduce the main results disclosed? [\[Yes\]](#) The protocol and appendix state splits, model dimensions, loss, optimizer, point counts, checkpoint policy, metrics, and seeds, and the artifact includes executable code and raw logs.
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6. **Experimental setting/details.** Are training and test details specified? [\[Yes\]](#) Section 4 and Appendix A enumerate them, and the frozen JSON plans give the machine-readable form.
7. **Experimental statistical significance.** Are error bars or appropriate uncertainty information reported? [\[Yes\]](#) All headline values include sample SD across three seeds; the predeclared decision rule requires a practical margin exceeding pooled seed variation, and constituent specifications are shown separately.
8. **Compute resources.** Are resources sufficient for reproduction reported? [\[Yes\]](#) The paper gives the device and software stack, and every training log records wall time, GPU, dtype, versions, and point/epoch counts.
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11. **Safeguards.** Are safeguards needed for released high-risk assets? [\[NA\]](#) No high-risk model or dataset is released.
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14. **Crowdsourcing and human subjects.** Are participant materials and compensation disclosed? [\[NA\]](#) No crowdsourcing or human subjects are involved.
15. **IRB approvals.** Are participant risks and approvals disclosed? [\[NA\]](#) No human-subject research is involved.
16. **Declaration of LLM usage.** Is important LLM use described? [\[Yes\]](#) An automated language-model coding agent assisted experiment design, implementation, execution, analysis, and manuscript preparation. The machine-readable plans, full raw traces, deterministic generators, and audits are provided so that every empirical statement can be independently checked.